

Effect of Numerical Integration on Critical Time Evaluation in Power System Stability Studies

Mikhail Y. Borodulin

System and Resource Planning
New York Independent System Operator (NYISO)
Rensselaer, New York, USA
mborodulin@nyiso.com

Abstract-- The effect of numerical integration on time-domain power system simulations when evaluating the fault critical clearing time (CCT) is discussed in relation to stability and accuracy of integration methods. The specificity of CCT simulations is that the power system is close to its physical stability limits. While decreasing the time step is widely thought to be an effective measure to prevent numerical instability and ensure sufficient accuracy, it is shown that in real stability studies, numerical phenomena can be non-monotonic and cause misrepresentation of power system transient stability. For a given method, the effect depends on a combination of the loading of the generator whose CCT is evaluated and fault and network characteristics. The effect is illustrated with real-world simulation results. The paper makes recommendations on how the quality of CCT evaluation can be enhanced.

Index Terms-- Numerical simulation, numerical stability, power system analysis, power system faults, power system stability.

I. INTRODUCTION

In power system stability analysis, evaluation of the fault critical clearing time (CCT) is intended to ensure that equipment capabilities (in particular, primary and backup protection systems' breaker characteristics) are adequate.

In this paper, the term "power system stability" is understood in accordance with definitions given in [1], [2]. The CCT is defined (for an individual generator and with respect to a particular fault) as the longest fault duration for which the generator and the power system remain transiently stable. Here, the term "particular fault" refers to a given fault scenario and given network conditions. Other CCT definitions, especially, for non-conventional generation (for example, for wind power plants [3]), and related aspects of power system dynamic performance are not considered.

Generally speaking, the problem of CCT evaluation has a mathematical formulation, which allows the use of direct methods based on energy functions or trajectory sensitivity analysis (examples and further references can be found in [4]-[7]). While such approaches may provide more insight into qualitative aspects of the problem, their applicability is limited to relatively small power systems.

In stability studies for large power systems, CCT evaluation is based on time-domain numerical simulations. No matter which technique is used to reduce the computational time, the evaluation essentially is a sequence of simulation runs with different fault clearing times.

The procedure of numerical integration of differential equations is well-known to be able to affect simulation results significantly; often, it creates an illusion of plausible processes while in fact they are nothing but numerical phenomena. For a single-machine system, [1] illustrates the effect of the integration step size on the rotor angle response for the 4th order Runge-Kutta methods (in Gill's modification) [1, p. 870] and for the trapezoidal rule [1, p. 872]. A variety of numerical phenomena observed when modeling large power systems with wind turbine generators is described in [8]. Experience with real system planning studies has suggested that CCT simulations also can be very sensitive to numerical integration.

This paper discusses the effect in relation to numerical stability and accuracy of integration methods. The specificity of CCT simulations is that the power system is close to its physical stability limits. While decreasing the time step is widely thought to be an effective measure to prevent numerical instability and ensure sufficient accuracy, it is shown that in real calculations, numerical phenomena can be non-monotonic and cause misrepresentation of power system transient stability. For a given integration method, the effect depends on a combination of the loading of the generator whose CCT is evaluated and fault and network characteristics.

The effect is illustrated with real-world simulation results. Calculations are performed using the Siemens PTI PSS/E dynamic program and real New York State power system databases applied in planning studies. To explain numerical phenomena, the paper refers to the approach [9] to evaluation of accuracy in terms of distortions of damping and frequency of free motion modes. Finally, recommendations are made on how the quality of CCT evaluation can be enhanced.

The work is part of New York Independent System Operator (NYISO) activities intended to ensure the high quality of power system stability studies.

II. GENERAL NOTES ON NUMERICAL SOLUTION IN POWER SYSTEM SIMULATION

This section outlines relevant features of a mathematical model of a power system and briefly discusses numerical solution of its equations in commercial dynamic programs.

A. Mathematical Model of a Power System

Usually, stability studies for large power systems are performed based on positive-sequence programs such as the GE Energy PSLF and Siemens PTI PSS/E. The idea of these tools is that for electrical machines, HVDC transmissions, FACTS devices and other components whose dynamics is intended to be reproduced, differential equations are integrated numerically, while the transmission network is represented by algebraic nodal equations.

Besides differential and algebraic equations, the mathematical model includes a variety of difference equations (in particular, describing discrete controls). More discrete functionality is added by external disturbances, protection relay operation and other discrete events. Therefore, a mathematical model of a power system is a complex nonlinear discrete-continuous dynamical system. The eigenvalues λ of the Jacobian matrix calculated at different time instants within intervals of continuity are important characteristics of the system dynamics.

B. Numerical Solution of Power System Equations

For numerical integration, both PSLF and PSS/E programs use the explicit two-step Adams-Bashforth (Adams-Bashforth-2) method with constant time step $\tau = \text{const}$. PSS/E has also a simulation mode based on an explicit scheme approximating the implicit trapezoidal rule [10], but it is not in a wide use. Some other commercial dynamic programs offer a set of integration methods.

Often, dynamic models of power system components (especially, non-conventional generation) include internal numerical solution algorithms. As a result, the process of integration of the overall differential system is multiply decomposed. Note that in PSS/E, the global integration procedure based on the Adams-Bashforth-2 method is used for most synchronous generators and their controls.

To solve algebraic network equations, a Newton-type iterative scheme is used. From experience with PSS/E-based studies, algebraic solution parameters, when in reasonable range, have little impact on CCT evaluation.

III. NUMERICAL INTEGRATION WHEN POWER SYSTEM IS PHYSICALLY UNSTABLE

When the fault clearing time T_C exceeds the CCT, the power system is physically unstable. Thus, the CCT evaluation at its core assumes simulations when the power system is close to physical stability limits and inevitably involves calculations for a physically unstable system.

In power system time-domain simulations, the term “numerical instability” is often used to characterize a situation when the process of numerical integration is unstable, no matter whether the power system is physically stable or not.

However, for a physically stable system, the process of numerical integration can be either stable or unstable; the same is for a physically unstable system.

This section follows the logic of [11] and discusses the applicability of the most popular methods of integration of ordinary differential equations for CCT simulations from the perspective of numerical stability and accuracy. Considered are one-step (Runge-Kutta) and linear multi-step methods. The section refers to their well-known stability regions in the complex plane $\lambda\tau$ and to the concept of A-stability [12], [13]. It is assumed that the integration is performed with $\tau = \text{const}$.

For one-step methods, for sake of brevity, stability functions will be described with reference to either the Taylor series $T_m(\lambda\tau)$ ($m = 1, 2, \dots$) for the exponential function $\exp(\lambda\tau)$ or to the Padé approximation $P_{m,n}(\lambda\tau)$ ($m = 0, 1, 2, \dots$ and $n = 0, 1, 2, \dots$) [12] to $\exp(\lambda\tau)$.

A. Applicability of One-Step Methods

The one-step explicit methods with stability functions $T_m(\lambda\tau)$ and $P_{m,0}(\lambda\tau)$ are not A-stable. Examples can be the forward Euler method $T_1(\lambda\tau) = P_{1,0}(\lambda\tau)$ or the 4th order Runge-Kutta method $T_4(\lambda\tau)$ (both its classical scheme and the Gill's modification). For these methods, the numerical stability border mostly lies in the left half-plane $\text{Re}(\lambda\tau) < 0$. This means that (with a few exceptions for multi-stage schemes) no matter what step τ is used, they do not produce a stable solution for a physically unstable system.

More specifically, explicit m -stage methods $T_m(\lambda\tau)$ with $m = 4$ and $m = 5$ (for example, the 4th order Runge-Kutta method and the Runge-Kutta-Merson method) are stable in some areas of the right half-plane $\text{Re}(\lambda\tau) > 0$ (for relatively large $\text{Im}(\lambda\tau)$ values). That is, these methods reproduce the corresponding physically unstable oscillatory modes as stable.

On the other hand, for a physically stable system and with a relatively large τ , when all $\lambda\tau$ points are in the left half-plane, the solution can be unstable but do not blow up, which creates an illusion of a physical instability. Therefore, while limiting the step τ , one-step explicit methods are suitable for CCT simulations; for relatively large τ , care should be taken of the most high-frequency and fast-damping modes.

For the A-stable one-step implicit methods with stability functions $P_{k,k}(\lambda\tau)$, the imaginary axis $j\text{Im}(\lambda\tau)$ is the numerical stability border (an example is the trapezoidal rule $P_{1,1}(\lambda\tau)$). The $P_{k,k}(\lambda\tau)$ methods are most suitable for CCT simulations from the perspective of numerical stability. Unfortunately, multi-stage schemes ($k \geq 2$) require considerable computational effort to calculate the time derivatives at each stage and step; as to the trapezoidal rule ($k = 1$), it can be effectively implemented for large systems.

The A-stable methods $P_{m,n}(\lambda\tau)$ with $m < n$ (in particular, the backward Euler method $P_{0,1}(\lambda\tau)$) are numerically stable in large areas of the right half-plane $\text{Re}(\lambda\tau) > 0$. That is, when some $\lambda\tau$ points belong to such areas and the system is physically unstable, the process of integration is stable no matter what step τ is used. Therefore, these methods are less applicable for CCT simulations. As an example, [17] indicates that the backward Euler method can mask physical instability.

B. Applicability of Linear Multi-Step Methods

The explicit Adams-Bashforth methods (with the number of steps $s \geq 2$) are superior to explicit one-step m -stage methods ($m = 4$ and $m = 5$) in terms of computational effort. However, they are not A-stable; usually, the stability region is smaller than those for one-step methods. Also, for $s = 3$, small numerically stable areas appear in the right half-plane.

For the Adams-Bashforth-2 method ($s = 2$), applied in both PSLF and PSS/E dynamic programs, some part of the numerical stability border nearly coincides with a segment of the imaginary axis $j\text{Im}(\lambda\tau)$ (say, on the interval $|\text{Im}(\lambda\tau)| < 0.3$), which provides nearly A-stable properties with respect to at least corresponding oscillatory modes of the free motion and makes this method suitable for CCT simulations.

The implicit Adams-Moulton methods with $s = 1$ and $s = 2$ coincide with the backward Euler method $P_{0,1}(\lambda\tau)$ and the trapezoidal rule $P_{1,1}(\lambda\tau)$, respectively; methods with $s > 2$ do not seem to be used in commercial power system analysis software. For other popular s -step ($s > 1$) schemes (such as the Gear methods or the Backward Differentiation Formulas), there are large areas of numerical stability in the right half-plane $\text{Re}(\lambda\tau) > 0$, which makes these methods less preferable.

C. Accounting for Accuracy of Methods

Sections IIIA and IIIB discussed the methods from the perspective of their numerical stability (and with respect to a linear dynamical system). However, when the process of integration is adequate in terms of numerical stability, the accuracy may be insufficient.

Even for a small but stiff linear dynamical system, integration can produce non-monotonic phenomena and other effects that find no explanation in terms of truncation or rounding error [8], [9]. Therefore, this paper refers to the approach [9], which evaluates the accuracy in terms of distortions of the damping and frequency of free motion modes. More detail on distortion characteristics for different integration schemes can be found in [14]–[16].

IV. ORGANIZATION OF NUMERICAL EXPERIMENTS

In numerical experiments intended to illustrate the effect, three synchronous generators with maximum active power output in the range of 300–600 MW and the pre-disturbance loading $P_0 \geq 75\%$ (of the maximum active power output) are monitored. These generators will be referred to as G1, G2 and G3 (generators G1 and G2 belong to the same power plant).

Considered are three different PSS/E dynamic setups representing the real NYISO system under different dispatch scenarios: they will be referred to as Network 1, 2 and 3 (in order of growth of the total generation). Applied are two three-phase, normally cleared faults (F1 and F2) on 345 kV buses (the fault F2 is more electrically distant).

For each experiment, phenomena are illustrated by a series of swing curves showing the evolution of the rotor angle θ for different values of the step τ . Normally, NYISO studies are performed with $\tau = 4.1670$ ms ($1/4$ of a 60 Hz frequency cycle). The clearing time T_C is expressed in 60 Hz cycles (c).

V. NUMERICAL PHENOMENA (SIMULATION RESULTS)

This section illustrates the effect of the global integration step τ in PSS/E simulations with the Adams-Bashforth-2 method. Fig. 1 though Fig. 4 demonstrate that for the same clearing time T_C , depending on the step τ , a simulation experiment can suggest either physical stability or instability exemplifying the extent to which numerical phenomena are non-monotonic. These figures also illustrate the role of generator loading P_0 and fault and network characteristics.

For example (see Fig. 1 for the fault F1 and generators G1 and G2 of the same power plant): $\tau = 4.1000$ ms and $\tau = 4.5000$ ms produce stable processes, while $\tau = 4.1670$ ms suggests instability. For the more distant fault F2 (see Fig. 2a): while $\tau = 3.3330$ ms, $\tau = 4.1678$ ms and $\tau = 4.7000$ ms produce stable processes, $\tau = 4.1673$ ms suggests instability. From Fig. 2: while for $T_C = 22$ c, the processes are unstable with $\tau = 4.1673$ ms, they are stable for $T_C = 23$ c with $\tau = 4.1670$ ms and $\tau = 4.4500$ ms. Note that for generators G1 and G2, the loading of either of them may affect the phenomena (and the CCT assessment) for the other one.

Fig. 3 shows a case where for generator G2, three different values – $\tau = 3.333$ ms, $\tau = 4.100$ ms and $\tau = 4.500$ ms – produce nearly identical stable results for $T_C = 16$ c, while for $T_C = 17$ c, the processes are stable for $\tau = 4.100$ ms but unstable for $\tau = 3.333$ ms and $\tau = 4.500$ ms. Fig. 4 in combination with Fig. 3a illustrates the role of network conditions (for $T_C = 16$ c): for Network 1 and Network 3, both $\tau = 3.333$ ms and τ around 4.100 ms and $\tau = 4.500$ ms produce stable results; however, for Network 2, τ around 4.100 ms suggests instability.

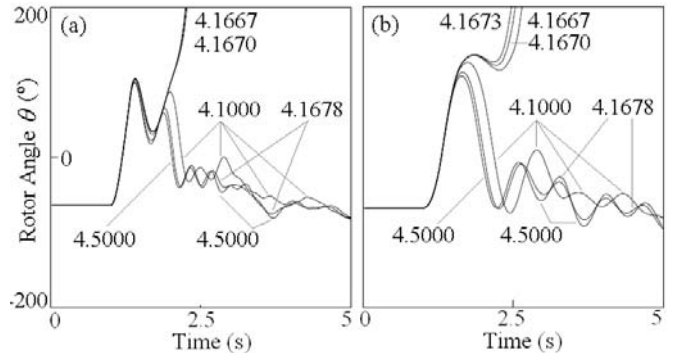


Fig. 1. Fault F1 simulations with different τ (ms), $T_C = 19$ c; Network 3: (a) – Generator G1 ($P_0 = 75\%$); (b) – Generator G2 ($P_0 = 80\%$).

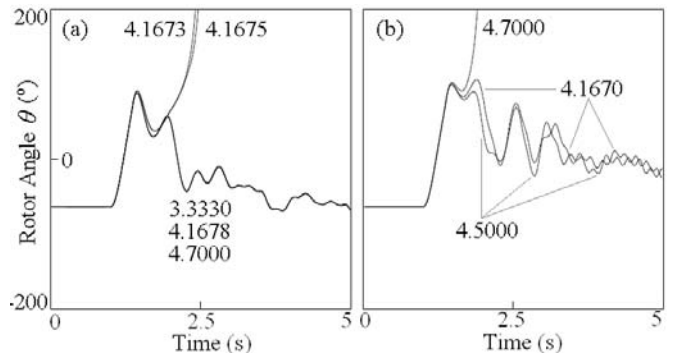


Fig. 2. Fault F2 simulations with different τ (ms), Generator G1 ($P_0 = 75\%$), Network 3: (a) – $T_C = 22$ c, (b) – $T_C = 23$ c.

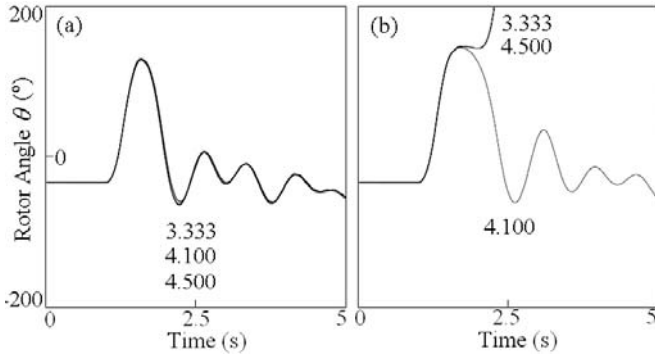


Fig. 3. Fault F1 simulations with different τ (ms) for Generator G2 ($P_0 = 96\%$), Network 1: (a) – $T_C = 16$ c, (b) – $T_C = 17$ c.

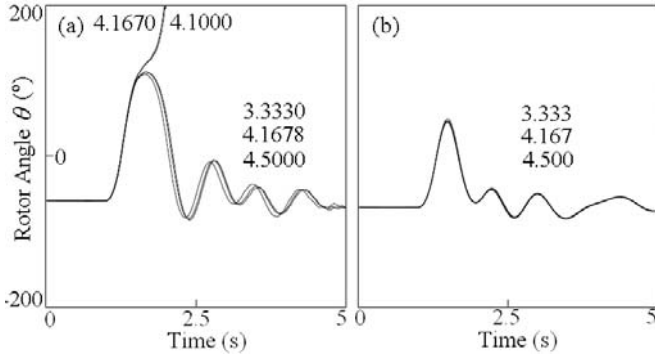


Fig. 4. Fault F1 simulations with different τ (ms) for Generator G2, $T_C = 16$ c: (a) – Network 2 (G2 loading $P_0=96\%$); (b) – Network 3 (G2 loading $P_0 = 99\%$).

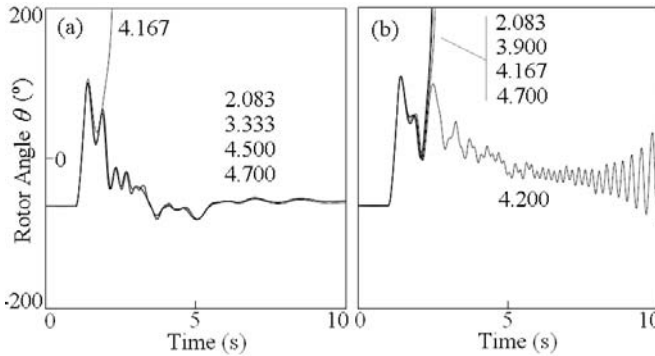


Fig. 5. Fault F1 simulations with different τ (ms) for Generator G1 ($P_0 = 75\%$), Network 3: (a) – $T_C = 19.0$ c; (b) – $T_C = 19.5$ c.

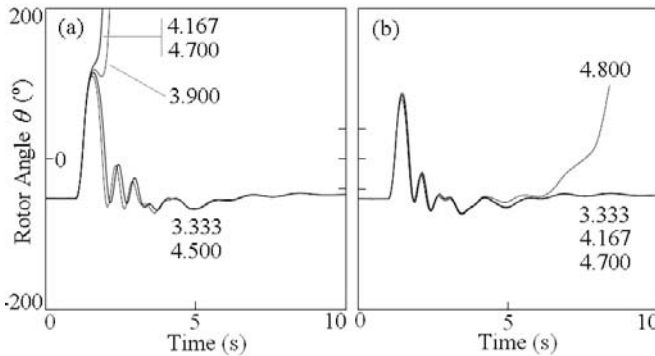


Fig. 6. Fault F1 simulations with different τ (ms) for Generator G3 ($P_0 = 100\%$), Network 3: (a) – $T_C = 15.5$ c; (b) – $T_C = 14.5$ c.

Fig. 1 through Fig. 4 show processes creating an illusion of a first-swing or second-swing physical instability that can be attributed to the lack of the synchronizing torque. Fig. 5 demonstrates that an illusion of a physical instability caused by the lack of the damping torque is also a possibility.

Fig. 6 shows non-monotonic phenomena for generator G3. Here, while $\tau = 4.167$ ms and $\tau = 4.700$ ms produce unstable processes for $T_C = 15.5$ c, the results are stable for $T_C = 14.5$ c. For this generator G3, Fig. 6b illustrates an experiment where a relatively large step $\tau = 4.800$ ms produces plausible results on the time interval of a few seconds while later there starts a process that presumably is a manifestation of a numerical instability phenomenon.

Swing curves of Fig. 1 through Fig. 6 illustrate processes for the monitored generators loaded at a level $P_0 \geq 75\%$. Other experiments have suggested that when the loading P_0 is lower, results are sensitive to the step τ more seldom; however, it can happen, too. In terms of other physical characteristics, neither the type of a generator nor its parameters nor features or parameters of its controls can be seen as factors that solely determine or dominate the effect of the step τ . With regard to the generator inertia constant, it is noteworthy that if, for the sake of experiment, it is artificially enlarged, the effect may become weaker. However, with real-world inertia constants, noticeable non-monotonic phenomena can well occur with both larger and smaller values.

It is clear that for a particular fault (in terms of its location, type and clearing procedure), network conditions (especially, in the vicinity of the monitored generator) can affect the picture of numerical phenomena considerably. However, none of these fault or network characteristics seems to be able to determine by itself the impact of τ , either. Also, there is no indication that insufficient tuning of control parameters or erroneous dynamics data, which cannot be fully avoided in real simulation environment, are factors of this kind.

It must be noted that in real power system studies, while the time step τ often affects the damping and frequency of essential modes considerably, a situation when a reasonable change in τ changes the reproduction of the generator rotor angle response from “stable” to “unstable” or vice versa is rather unusual. Nevertheless, non-monotonic phenomena related to τ are no exception, either; when they occur, they can misrepresent the power system transient stability. Based on numerous simulation experiments, it is a combination of the generator loading and fault and network characteristics that determines the picture of numerical phenomena. For some considered databases and faults, depending on how CCT simulations are performed, a possible difference in the CCT assessments with different τ can be 1 to 2 cycles.

VI. DISCUSSION OF RESULTS

Based on [9], [14]–[16], for many integration methods, damping and frequency distortions in certain areas of the complex plane $\lambda\tau$ along the imaginary axis $\text{Im}(\lambda\tau)$ are such that a relatively small change in the step τ can affect simulation results considerably. For a large stiff nonlinear system, the picture of numerical phenomena can be really complex, especially when multi-step methods are applied.

For a generator subjected to a disturbance, the CCT (as defined) is a transient stability limit determined by either the lack of the synchronizing torque or the lack of the damping torque. It is well-known that the lack of the synchronizing torque leads to aperiodic instability, while the lack of the damping torque leads to oscillatory instability [2]. The graphs included in Section V may suggest that high sensitivity to the step τ , when it is the case, is a result of a large damping distortion respectively for a relevant aperiodic ($\text{Im}(\lambda) = 0$) or oscillatory ($\text{Im}(\lambda) \neq 0$) mode created by the subject generator. However, the simulations have not provided sufficient evidence that this straightforward mechanism determines the effect of τ : it is more likely that high-sensitivity and non-monotonic phenomena are a result of simultaneous distortions of a number of essential solution modes, aperiodic and oscillatory, related not only to the subject generator.

Generally speaking, most phenomena described in Section V can be seen as a manifestation of the following interrelated effects: inadequate reproduction of the damping and/or frequency for some essential modes; numerical instability with a smaller step τ ; a more accurate solution with a larger step τ ; and excessive sensitivity to the step τ .

Hypothetically, each specific phenomenon of this kind can be explained using the approach [9]. However, due to dimensionality, complexity and variability of a large power system, such investigations would be very time-consuming; their results would be case-specific so that a new investigation would be needed for each new questionable phenomenon. On the other hand, in real system panning studies, simply carrying out additional simulations with different values of the step τ and, when available, applying integration methods belonging to different families seems to be sufficient to eliminate the uncertainty caused by non-monotonic phenomena.

As indicated, Section V results illustrate the effect of τ in PSS/E simulations with the Adams-Bashforth-2 method. Due to a great deal of resemblance between PSS/E and PSLF, the observations are relevant to PSLF simulations, too. For these two programs, it is recommended that a questionable fault simulation with the clearing time T_C close to the supposed CCT value be repeated using different τ values, both larger and smaller than those normally applied, within a representative range allowing an understanding of how the step affects the assessment of physical stability.

If a dynamic program is equipped with a set of integration methods, it is of great help in making CCT evaluation more efficient. From a general perspective, when several methods are available, it is recommended that a questionable simulation with T_C close to the CCT be repeated using methods with significantly different stability and accuracy properties with respect to solution modes near the imaginary axis $\text{Im}(\lambda\tau)$ (say, using a combination of an explicit and implicit scheme). However, further studies are needed to analyze the applicability and effectiveness of this approach in large power system studies. Also, when possible, it is desirable that the organization of the numerical integration be coordinated with an *a priori* understanding of how a physical instability would develop (in terms of $\text{Re}(\lambda)$ and $\text{Im}(\lambda)$ ranges for essential free motion modes).

VII. CONCLUSIONS

In power system stability studies, when the fault clearing time is close to the CCT, numerical integration can be very sensitive to the time step and cause misrepresentation of transient stability. Due to non-monotonic numerical phenomena, decreasing the time step does not necessarily ensure a more accurate solution. Simulations with different steps and/or using methods with different properties (with respect to solution modes near the border of numerical stability) are needed to enhance the quality of CCT evaluation.

REFERENCES

- [1] P. Kundur, *Power System Stability and Control*. New York: McGraw-Hill, Inc., 1994.
- [2] P. Kundur, J. Paserba, V. Ajjarapu et al., "Definition and classification of power system stability," *IEEE Trans. on Power Systems*, Vol. 19, No. 2, May 2004.
- [3] A. Adamczyk, R. Teodorescu, R.N. Mukerjee, and P. Rodriguez, "Overview of FACTS devices for wind power plants directly connected to the transmission network," in *Proc. of IEEE Int. Symposium on Industrial Electronics (ISIE 2010)*, ISIE Press, 2010, p. 3742-3748.
- [4] A. Llamas, J. De La Ree Lopez, L. Milli, A.G. Phadke, and J.S. Thorp, "Clarifications of the BSU method for transient stability analysis," *IEEE Trans. on Power Systems*, Vol. 10, No. 1, Feb. 1995.
- [5] T.B. Nguyen, M.A. Pai, and I.A. Hiskens, "Computation of critical values of parameters in power systems using trajectory sensitivities," in *Proc. of the 14th Power Systems Computational Conference (PSCC'02)*, June 24-28, 2002, Seville, Spain.
- [6] Y. Tada, T. Takazawa, H.-D. Chiang, H. Li, and J. Tong, "Transient stability evaluation of a 12,000-bus power system data using TEPCO-BCU," in *Proc. of the 15th Power Systems Computational Conference (PSCC'05)*, Aug. 22-26, 2005, Liege, Belgium.
- [7] A. Hoballah and I. Erlich, "Real-time operation of deregulated electricity market: An integrated approach to dynamic stability assurance," in *Proc. of 2010 IEEE Int. Energy Conference and Exhibition (EnergyCon)*, Dec. 18-22, 2010, Manama, Bahrain.
- [8] M. Y. Borodulin, "Numerical effects in stability simulations of power systems with wind turbine generators," in *Proc. of 2012 IEEE PES General Meeting*, July 22-26, 2012, San Diego, CA, USA.
- [9] M. Borodulin, "An approach to evaluating accuracy of numerical integration in simulation of linear network transients," in *Proc. of the 14th Power Systems Computational Conference (PSCC'02)*, June 24-28, 2002, Seville, Spain.
- [10] F.P. De Mello, J.W. Feltes, T.W. Laskowski et al., "Simulating fast and slow dynamic effects in power systems," *IEEE Computer Applications in Power*, Vol. 5, Issue 3, 1992, p. 33-38.
- [11] M. Y. Borodulin, "Digital simulation of unstable dynamical systems," *Izv. AN. Energetika i Transport*, 1991, No. 3, p.101-110 (in Russian).
- [12] K. Dekker and J.G. Verwer, *Stability of Runge-Kutta Methods for Stiff Non-Linear Differential Equations*. Amsterdam: North-Holland, 1984.
- [13] L.O. Chua and P.-M. Lin, *Computer-aided analysis of electronic circuits*. Englewood Cliffs, New Jersey: Prentice-Hill, 1975.
- [14] M.Y. Borodulin, "Filtering oscillatory modes of the general solution when numerically integrating differential equations of linear electrical circuits," *Elektrichestvo*, 1990, No. 11, p. 49-54 (in Russian).
- [15] M.Y. Borodulin, "Distortions of aperiodic modes of the general solution when digitally simulating transients in linear electrical circuits," *Elektrichestvo*, 1991, No. 7, p. 41-46 (in Russian).
- [16] M.Y. Borodulin, "Accuracy of multi-step integration methods when calculating transients in electrical circuits," *Elektrichestvo*, 1993, No. 7, p. 57-63 (in Russian).
- [17] C. A. Felippa and K. C. Park, *Synthesis tools for structural dynamics and partitioned analysis of coupled systems*. Chapter in *Engineering Structures under Extreme Conditions*, ed. by A. Ibrahimbegovic and B. Brank. Amsterdam: IOS Press, 2006.